

CBSE Practice Worksheet

Class 7 | Mathematics | Difficulty: Hard | Set B

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. If $\log_{\blacksquare} 8 = x$, find x
(A) 2
(B) 3
(C) 4
(D) 8
2. Factorize: $x^2 - 5x - 14$
(A) $(x-7)(x+2)$
(B) $(x+7)(x-2)$
(C) $(x-7)(x-2)$
(D) $(x+7)(x+2)$
3. What is the cube root of -125?
(A) 5
(B) -5
(C) ± 5
(D) undefined
4. Solve: $3(x-2)/4 = (2x+1)/3$
(A) $x = 14$
(B) $x = 22$
(C) $x = 26$
(D) $x = 30$
5. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
(A) $x - 2$
(B) $x + 2$
(C) $(x-2)^2$
(D) $x^2 - 4$
6. If $2\blacksquare = 3\blacksquare = 6\blacksquare$, which is true?
(A) $1/z = 1/x + 1/y$
(B) $z = x + y$
(C) $z = xy$
(D) $z = x - y$
7. If $x + 1/x = 5$, find $x^2 + 1/x^2$
(A) 21
(B) 23
(C) 25
(D) 27
8. If $a:b = 3:4$ and $b:c = 5:6$, find $a:c$
(A) 3:6
(B) 5:8
(C) 15:24

(D) 1:2

9. Find: $(-1)^1 \blacksquare + (-1) \blacksquare$

(A) 0

(B) 2

(C) -2

(D) 1

10. Solve: $|2x - 5| = 9$

(A) $x = 7$ or -2

(B) $x = 7$ or 2

(C) $x = -7$ or 2

(D) $x = -7$ or -2

Section B: Fill in the Blanks (1 mark each)

11. HCF $\times$ LCM of 12 and 18 = ____

12. If $x^2 - 6x + 8 = 0$, then $x =$ ____ or ____

13. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/$ ____

14. $(x + 1/x)^2 - (x - 1/x)^2 =$ ____

15. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 -$ ____

16. $\sqrt[3]{-64} =$ ____

17. If $a + b + c = 0$, then $a^3 + b^3 + c^3 =$ ____

18. If $5^x = 125$, then $x =$ ____

Section C: Short Answer Questions (2 marks each)

19. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

20. Find the value of x : $3^{(2x-1)} = 27^{(x-2)}$

21. Factorize: $8x^3 - 27$

22. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

23. Solve: $|x - 3| + |x + 2| = 7$

24. Solve: $x^2 - 7x + 12 = 0$

25. Solve: $\log \blacksquare(x - 1) + \log \blacksquare(x + 1) = 3$

--- End of Worksheet ---

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